CHECK WHAT YOU LEARNT

Check Yourself - Unit 5

A. Choose the correct answer

1. An IR sensor distinguishes black from white by measuring
   (a) temperature (b) reflected infrared light (c) sound (d) voltage from the floor
2. A black surface makes the sensor output
   (a) LOW (b) HIGH (c) nothing (d) 5 volts
3. The blue trimmer on an IR module adjusts
   (a) brightness (b) the comparator threshold (c) the speed (d) the colour
4. An H-bridge lets a circuit
   (a) measure distance (b) reverse a motor (c) store charge (d) read a sensor
5. The wire most often forgotten is
   (a) IR OUT to D2 (b) Arduino GND to L298N GND (c) the LED wire (d) the USB cable
6. A system whose own output changes its next input is called
   (a) an open loop (b) a closed feedback loop (c) a latch (d) a comparator

B. Fill in the blanks

1 An IR module contains an infrared ______ and a ______________.

2 IR sensors should sit about ______ to ______ millimetres above the floor.

3 Both sensors reading black usually means a ______________.

4 If the robot wobbles violently, the ______________ is probably too high.

C. Answer in one or two lines

1 Explain why an IR sensor works equally well in a dark room and a bright one.

2 Write out all four sensor combinations and the action each one requires.

3 Explain what a closed feedback loop is, using your robot as the example.

4 Which of the three projects this year taught you the most, and why?

Extra Challenges

Finished early? Pick any of these. They all use parts you already have.

#	Challenge	Uses
1	A two-stage fire alarm: warning first, full alarm on confirmation	flame + DHT11
2	An alarm that shows how many times it has triggered today	LCD + counter
3	A night-mode alarm that flashes silently after dark	LDR + buzzer
4	A scrolling message on the LCD, one character at a time	LCD + millis()
5	A custom character of your own design on the display	createChar()
6	A comfort index combining temperature and humidity into one number	DHT11 + maths
7	A robot that counts junctions as it follows the line	2 IR + counter
8	A robot that stops for 3 seconds at each junction, then continues	2 IR + millis()
9	A robot with three IR sensors that steers proportionally	3rd IR module
10	A line follower that shows its speed on the LCD as it drives	LCD + robot

MIXED REVISION

Twenty Questions to Test Yourself

1. State Ohm's law and explain each symbol.
2. Which chip on the Arduino Uno is the actual computer?
3. What does the 16 MHz crystal do?
4. Where does your compiled program live, and where do variables live?
5. Describe what the bootloader does.
6. How many different values can a 10-bit ADC report?
7. Work out the size of one ADC step across a 5 volt range.
8. What does a flame sensor actually detect?
9. Why can sunlight fool a flame sensor?
10. Explain why two sensors must agree before your alarm sounds.
11. What does it mean for an alarm to latch?
12. Why does a push switch need a pull-down resistor?
13. Explain the difference between a false positive and a false negative.
14. How does applying a voltage make an LCD dot go dark?
15. Why is 4-bit mode used to drive the display?
16. What does setCursor(0, 1) do?
17. Why is millis() better than delay() for rotating screens?
18. How does an IR sensor tell black from white?
19. Why must the Arduino and the L298N share a ground?
20. Name one improvement you would make to your robot with one more session.

Wiring Reference - All Three Projects

Keep this page open while you build. Every pin used this year, in one place.

Project 1 - Fire Security Alarm

Arduino pin	Connects to
D2	Flame sensor DO
A0	Flame sensor AO
D4	DHT11 DATA
D7	Push switch, with 10K to GND
D8	Buzzer +
D9	Orange LED through 220 ohm
D10	Blue LED through 220 ohm
5V and GND	All module VCC and GND

Project 2 - LCD Weather System

Arduino pin	Connects to
D12	LCD RS
D11	LCD E
D5, D4, D3, D2	LCD D4, D5, D6, D7 in that order
D6	DHT11 DATA
A0	LDR module AO
-	LCD V0 to the potentiometer wiper
-	LCD VSS and RW to GND, VDD to 5V
-	LCD A to 5V through 220 ohm, K to GND

Project 3 - Line Following Robot

Arduino pin	Connects to
D2	Left IR sensor OUT
D3	Right IR sensor OUT
D8, D7	L298N IN1, IN2 - left motor direction
D6, D5	L298N IN3, IN4 - right motor direction
D10, D9	L298N ENA, ENB - speed
GND	L298N GND - the shared ground, never omit
-	L298N 12V and GND to the battery pack
-	L298N OUT1-4 to the two motors

Words to Know

Word	Meaning
ADC	The circuit that turns a voltage into a number.
Analog	A value that can be anything within a range.
ATmega328P	The microcontroller chip that is the real computer on the board.
Backlight	The LED behind an LCD that supplies all the light you see.
Boolean	A variable that holds only true or false.
Bootloader	A built-in program that receives your code and writes it to flash.
Closed feedback loop	A system whose own output changes its next input.
Comparator	A circuit that flips its output when an input crosses a set level.
Debounce	Ignoring the brief chatter of a switch as it makes contact.
EEPROM	Small memory that keeps data after the power is removed.
False positive	Alarming when there is nothing to alarm about.
False negative	Staying silent when there really is something.
Flash memory	Where your compiled program is stored.
Floating input	An unconnected pin, reading randomly because of stray noise.
H-bridge	Four switches that let a circuit reverse a motor.
HD44780	The standard controller chip on a character LCD.
Infrared	Light just beyond red, invisible to us.
Latching	Staying on once triggered, until deliberately cleared.
Library	Ready-made code you include and use.
Liquid crystal	A state of matter that flows yet aligns like a crystal.
millis()	How long the board has been running, in milliseconds.
Non-blocking	Code that waits without freezing everything else.
Ohm's law	V = I x R.
Photodiode	A component that produces current when light falls on it.
Phototransistor	A component that conducts more when light falls on it.
Pull-down resistor	A resistor holding an input at 0V until something pulls it high.
Resolution	The smallest change a measuring system can detect.
Shared ground	The wire letting two supplies agree on what zero volts means.
SRAM	Memory holding every variable while the program runs.
System	Several parts working together, where the whole exceeds the parts.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one good answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(b)	V = I x R
2	(b)	13.6 mA
3	(a)	columns of five
4	(b)	a coded sequence of pulses
5	(b)	hold the input at a known level
6	(b)	has several parts working together

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	the ATmega328P chip
2	(b)	the clock
3	(b)	flash memory
4	(c)	1024 values
5	(a)	4.9 millivolts
6	(b)	receive your code and write it to flash

Check Yourself - Unit 3

Q	Answer	The correct option
1	(c)	infrared light
2	(b)	an optical filter
3	(b)	reduce false alarms
4	(b)	a human resets it
5	(b)	floating
6	(b)	false negative

Check Yourself - Unit 4

Q	Answer	The correct option
1	(b)	no light of its own
2	(b)	the potentiometer on V0
3	(b)	the display's controller chip
4	(b)	save Arduino pins
5	(b)	the last character of the bottom row
6	(b)	never freezes

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	reflected infrared light
2	(b)	HIGH
3	(b)	the comparator threshold
4	(b)	reverse a motor
5	(b)	Arduino GND to L298N GND
6	(b)	a closed feedback loop

My Project Log

Fill one row after every class. This is your evidence, and it counts towards your assessment.

Session	What I built or learnt	A problem I hit	How I fixed it
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

This year you stopped treating components as magic boxes. You know what is inside the chip, what an ADC really measures, why a flame sensor is not a thermometer, and why an LCD needs a light behind it. That habit - always asking how the thing actually works - is worth far more than any single project in this book.